I.1

$$A^T = \begin{pmatrix} 2 & 3 & -1 \\ -4 & 5 & 0 \end{pmatrix} \quad B^T = \begin{pmatrix} 1 & -3 & 5 \\ 2 & -4 & 2 \\ 7 & 0 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 6 & -3 & 9 \\ 4 & -5 & 2 \\ 8 & 1 & 5 \end{pmatrix}$$

$$A^T C = \begin{pmatrix} 12+12-8 & -6-15-1 & 18+6-5 \\ -24+20 & +12-25 & -36+10 \end{pmatrix} = \begin{pmatrix} 16 & -22 & 19 \\ -4 & -13 & -26 \end{pmatrix}$$

$$A^T B^T = \begin{pmatrix} 2+6-7 & -6-12 & 10+6-1 \\ -4+10 & 12-20 & -20+10 \end{pmatrix} = \begin{pmatrix} 1 & -18 & 15 \\ 6 & -8 & -10 \end{pmatrix}$$

$$D = \begin{pmatrix} 16-2 & -22+36 & 19-30 \\ -4-12 & -13+16 & -26+20 \end{pmatrix} = \begin{pmatrix} 14 & 14 & -11 \\ -16 & 3 & -6 \end{pmatrix}$$

I.2

$3x+2=8 \qquad x=2$

$3y-12=6 \qquad y=6$

$12+2z=4 \qquad z=-4$

$6+4=v \qquad v=10$

I.3

$\begin{cases} 1/3 = 3/p \\ 1/3 = 3/p \end{cases} \quad p=9 \qquad \begin{cases} 1/5 = 3/q \\ 1/3 = 5/q \end{cases} \quad q=15$

I.4

$$\begin{cases} 2\lambda_1 - \lambda_2 = 0 \\ -5\lambda_1 + 3\lambda_2 = 0 \end{cases} \qquad \begin{cases} 2\lambda_1 - \lambda_2 = 0 \\ -5\lambda_1 \cancel{+3\lambda_2} + 6\lambda_1 \cancel{-3\lambda_2} = 0 \end{cases} \qquad \begin{cases} \lambda_2 = 0 \\ \lambda_1 = 0 \end{cases}$$

вектора a_1 и a_2 линейно независимы

$$\begin{cases} 2x - y = 1 \\ -5x + 3y = -4 \end{cases} \qquad \begin{cases} 2x - y = 1 \\ -5x + 3y + 6x - 3y = -4 + 3 \end{cases} \qquad \begin{cases} y = -3 \\ x = -1 \end{cases} \quad [x]_B = \begin{pmatrix} -1 \\ -3 \end{pmatrix}$$

$$y = \begin{pmatrix} 2 \\ -5 \end{pmatrix} + \begin{pmatrix} -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

II.1

$$\nabla f(x) = \begin{pmatrix} 3x_1^2 - 2x_2 - 3 \\ -2x_1 + 2x_2 - 2 \end{pmatrix}$$

$$\begin{cases} 3x_1^2 - 2x_2 - 3 = 0 \\ -2x_1 + 2x_2 - 2 = 0 \end{cases} \quad \begin{cases} 3x_1^2 - 2x_1 - 5 = 0 \\ -x_1 + x_2 - 1 = 0 \end{cases} \quad \begin{cases} (x_1+1)(3x_1-5) = 0 \\ x_2 = x_1 + 1 \end{cases}$$

Критические точки: $\begin{cases} x_1 = -1 \\ x_2 = 0 \end{cases}$ $\quad \begin{cases} x_1 = 5/3 \\ x_2 = 8/3 \end{cases}$

II.2

$$\frac{\partial f}{\partial x_1} = \frac{1}{\sqrt{x_1} + \sqrt{x_2}} \cdot \frac{1}{2\sqrt{x_1}} \qquad \frac{\partial f}{\partial x_2} = \frac{1}{\sqrt{x_1} + \sqrt{x_2}} \cdot \frac{1}{2\sqrt{x_2}}$$

$$x_1 \cdot \frac{1}{\sqrt{x_1} + \sqrt{x_2}} \cdot \frac{1}{2\sqrt{x_1}} + x_2 \cdot \frac{1}{\sqrt{x_1} + \sqrt{x_2}} \cdot \frac{1}{2\sqrt{x_2}} =$$

$$= \frac{1}{2} \cdot \frac{1}{\sqrt{x_1} + \sqrt{x_2}} \left(\frac{x_1}{\sqrt{x_1}} + \frac{x_2}{\sqrt{x_2}} \right) = \frac{1}{2} \cdot \frac{\sqrt{x_1} + \sqrt{x_2}}{\sqrt{x_1} + \sqrt{x_2}} = \frac{1}{2}$$

III.3

$$J_f = \begin{pmatrix} 1 & 1 & 1 \\ yz & xz & xy \end{pmatrix} \qquad J_f(\bar{v}) = \begin{pmatrix} 1 & 1 & 1 \\ 6 & 3 & 2 \end{pmatrix}$$

II.4

$$\frac{\partial}{\partial x_i}\left(\frac{1}{3}\left(\sqrt{x_1^2+\ldots x_n^2}\right)^3\right)=$$

$$=\frac{1}{\cancel{3}}\cdot\cancel{3}\cdot\left(\sqrt{x_1^2+\ldots+x_n^2}\right)^{\cancel{2}}\cdot\frac{1}{\cancel{2}}\cdot\cancel{\frac{1}{\sqrt{x_1^2+\ldots+x_n^2}}}\cdot\cancel{2}x_i=\|x\|_2\cdot x_i$$

$$f'(x)=\|x\|_2\cdot\|x\|_1$$

$$\nabla f(x)=(\|x\|_2\cdot x_1,\ldots,\|x\|_2\cdot x_n)$$

II.5

$$\frac{\partial}{\partial x_i}\left(\sqrt{x_1^2+\ldots+x_n^2}\right)=\frac{1}{2}\cdot\frac{1}{\sqrt{x_1^2+\ldots+x_n^2}}\cdot 2x_i$$

$$f'(x)=\frac{\|x\|_1}{\|x\|_2}\qquad \nabla f(x)=\left(\frac{x_1}{\|x\|_2},\ldots,\frac{x_n}{\|x\|_2}\right)$$

II.6

$y = Ax \qquad y_i = (Ax)_i = A_{i1}x_1 + \ldots + A_{in}x_n \qquad \frac{\partial y_i}{\partial x_j} = A_{ij}$

$$\frac{\partial}{\partial x_j}\left(\sqrt{y_1^2 + \ldots + y_m^2}\right) = \frac{1}{2} \cdot \frac{}{\sqrt{y_1^2 + \ldots + y_m^2}} \cdot (2y_1 A_{1j} + \ldots + 2y_m A_{mj}) =$$

$$= \frac{(Ax)_1 A_{1j} + \ldots + (Ax)_m A_{mj}}{\|Ax\|_2}$$

$$f'(x) = \sum_j^m \frac{(Ax)_1 A_{1j} + (Ax)_m A_{mj}}{\|Ax\|_2}$$

$$\nabla f(x) = \left(\frac{(Ax)_1 \cdot A_{11} + \ldots + (Ax)_m A_{m1}}{\|Ax\|_2}, \ldots, \frac{(Ax)_n A_{1n} + \ldots + (Ax)_m \cdot A_{mn}}{\|Ax\|_2}\right)$$

II.7

$x \in \mathbb{R}^{n\times 1} \Rightarrow x^T \in \mathbb{R}^{1\times n}$

$x^T \cdot x = \sum_i^n x_i \cdot x_i = x_1^2 + \ldots + x_n^2$

$$\frac{\partial}{\partial x_i}\left(-e^{-(x_1^2 + \ldots + x_n^2)}\right) = -e^{-(x_1^2 + \ldots x_n^2)} \cdot (-2) \cdot x_i$$

$$f'(x) = 2e^{-\|x\|_2^2} \cdot \|x\|_1$$

$$\nabla f(x) = (2e^{-\|x\|_2^2} \cdot x_1, \ldots, 2e^{-\|x\|_2^2} \cdot x_n)$$